

# WASP-11b

R-1462

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-10\_141109 · created 2026-08-01T10:54:27+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456970.83832 +/- 0.00073 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1307 +/- 0.0075                 |
| Transit depth (Rp/Rs)^2                           | 1.71 +/- 0.2 [%]                  |
| Orbital Inclination (inc)                         | 89.97 +/- 0.5                     |
| Airmass coefficient 1 (a1)                        | 1.1401 +/- 0.0097                 |
| Airmass coefficient 2 (a2)                        | -0.121 +/- 0.015                  |
| Scatter in the residuals of the lightcurve fit is | 0.85 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 4.2                               |
| Optimal Annulus                                   | 15.75                             |
| Transit Duration (day)                            | 0.1067 +/- 0.001                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1307 $\pm$ 0.0075 vs 0.131 (deviation 0.04 $\sigma$ ) |
| Duration fit vs published            | 0.1067 d vs 0.1075 d (deviation 0.75 $\sigma$ )         |
| Mid-time O-C                         | -2.5 min                                                |
| Depth SNR                            | 16.3                                                    |
| Residual scatter                     | 0.85 %                                                  |

### Light curve

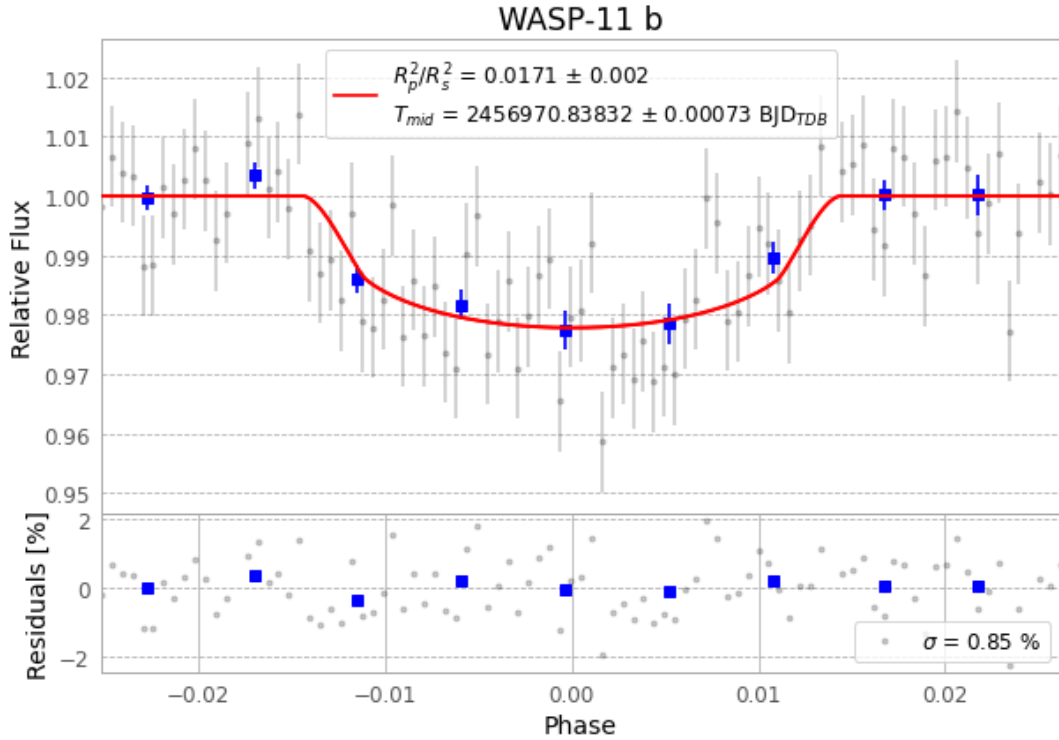


Figure 1 — FinalLightCurve\_WASP-11 b\_2014-11-08.png (SHA-256 2bd613af495822c5...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [519, 224], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 daed9b7c76a5d5a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

93 FITS frames (61 MB), HATP-10141109054212.FITS → HATP-10141109101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-141109.log (1624e4f3a67e...), FinalLightCurve\_WASP-11 b\_2014-11-08.png (2bd613af4958...), FinalLightCurve\_WASP-11 b\_2014-11-08.csv (0c1752965ccf...)

## Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)